

Whether the futures market is in contango or backwardated depends, among other things, on the usual elements of supply and demand plus the cost of storage, inventory imbalances, and *convenience yield*. The latter refers to the reward producers receive for storing a commodity, which investors do not receive: being able to keep production on schedule, the ability to profit from increasing production at short notice, and the ability to profit from temporary shortages in the commodity.

Returns on commodities depend crucially on the behavior of long-dated futures, so commodity investors need to keep an eye on the shape of the futures curve. In Table 11.7, the negative roll return in the 2000s was due to highly backwardated oil markets in 2008. The movement of the oil futures from contango in 2007 to backwardated in 2008, causing negative roll returns, was a double whammy for many investors who were already hurt by falling oil prices (negative spot returns) during that period. Most other commodity markets at that time were in contango.

Investors expecting positive commodity returns when commodity prices are increasing in spot markets will be in for a nasty surprise when their investments in futures turn out to be quite different from the spot returns. The United States Oil Fund is one of the largest exchange-traded funds (ETFs) with approximately \$1.3 billion net assets as of July 2012 (see chapter 16 for details on 40-Act funds) and is designed to track the price of oil.

The USO doesn't hold physical oil: it holds oil futures and is forced to roll every thirty days. Figure 11.9 plots the one-year moving average of USO returns versus crude oil spot returns from April 2007 to December 2012. There are pronounced discrepancies between the crude oil spot return and the USO return; on

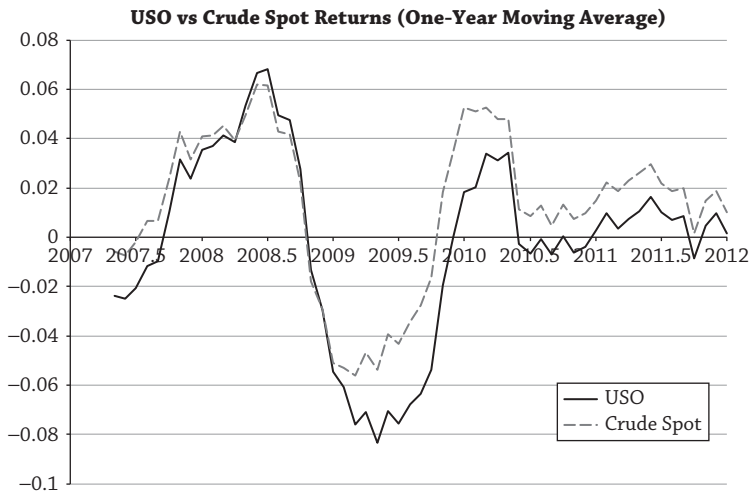


Figure 11.9